

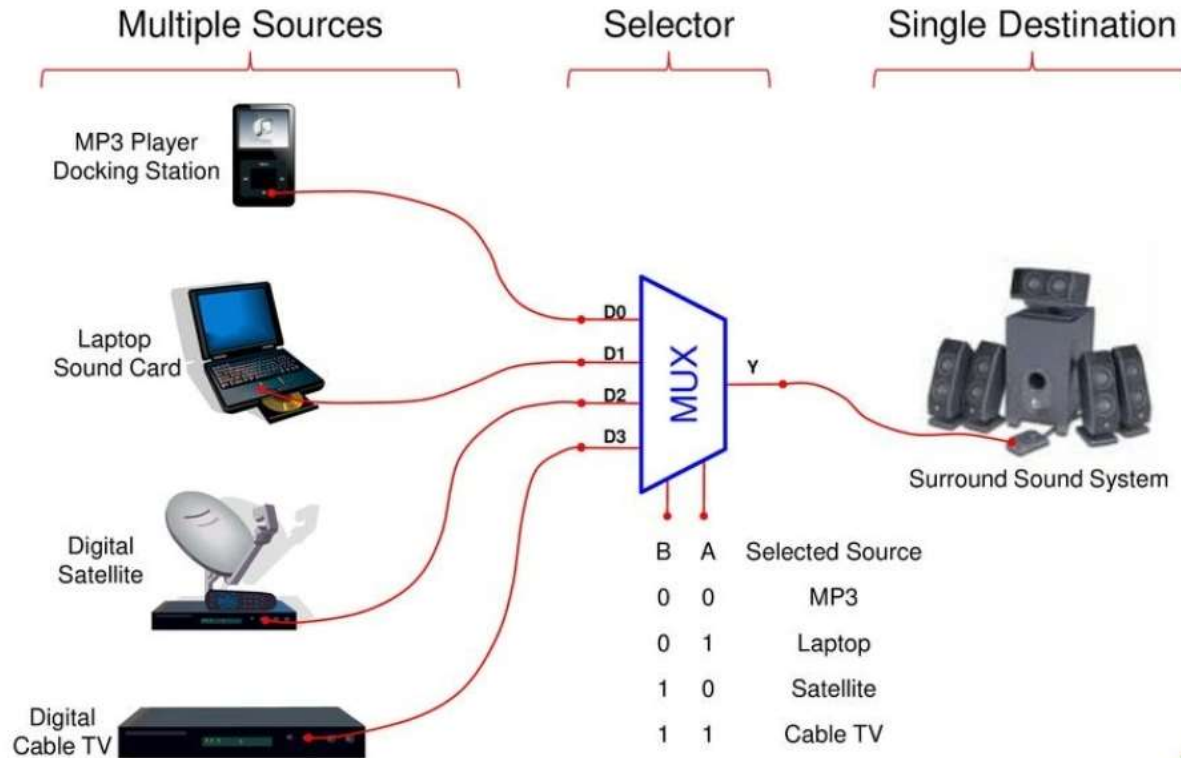
Multiplexer (Data Selectors)

- The term Multiplexer means many into one.
- Multiplexing is the process of transmitting a large number of information over a single line.
- A Digital Multiplexer (MUX) is a combinational Circuit that select one digital information from several sources and transmits the selected information on a single output line.
- A Multiplexer is also called a Data Selector.
- The Multiplexer has several data input line and a single output line.

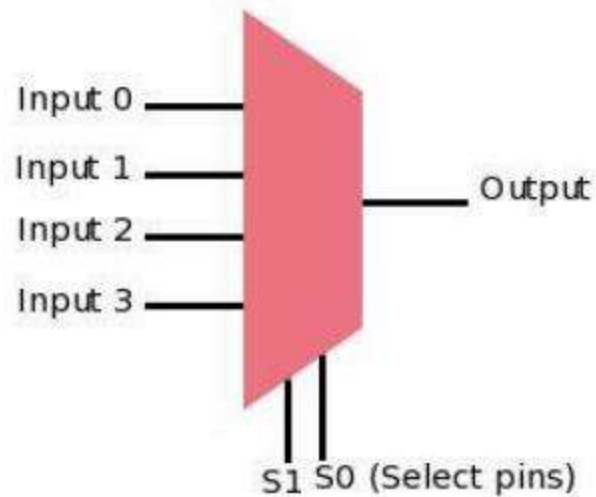
Cont.

- MUX directs one of the inputs to its output line by using a control bit word (*selection line*) to its select lines.
- Multiplexer contains the followings:
 - ❖ data inputs
 - ❖ selection inputs
 - ❖ a *single output*
- Selection input determines the input that should be connected to the output.
- The multiplexer acts like an electronic switch that selects one from different.

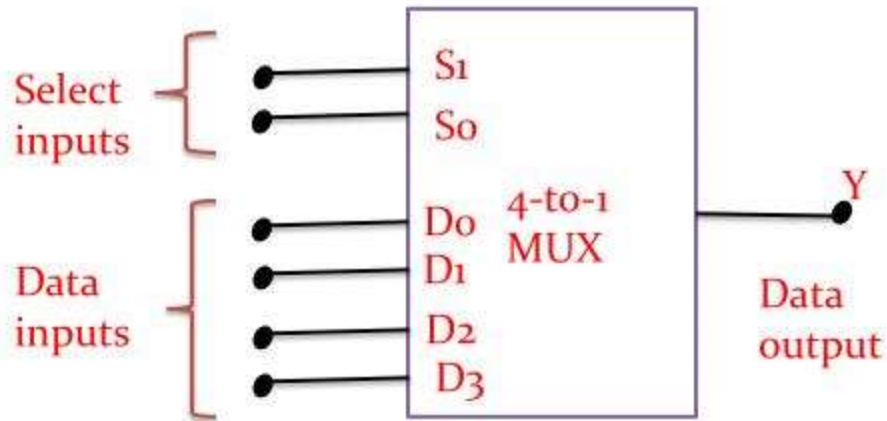
Typical Application of a MUX



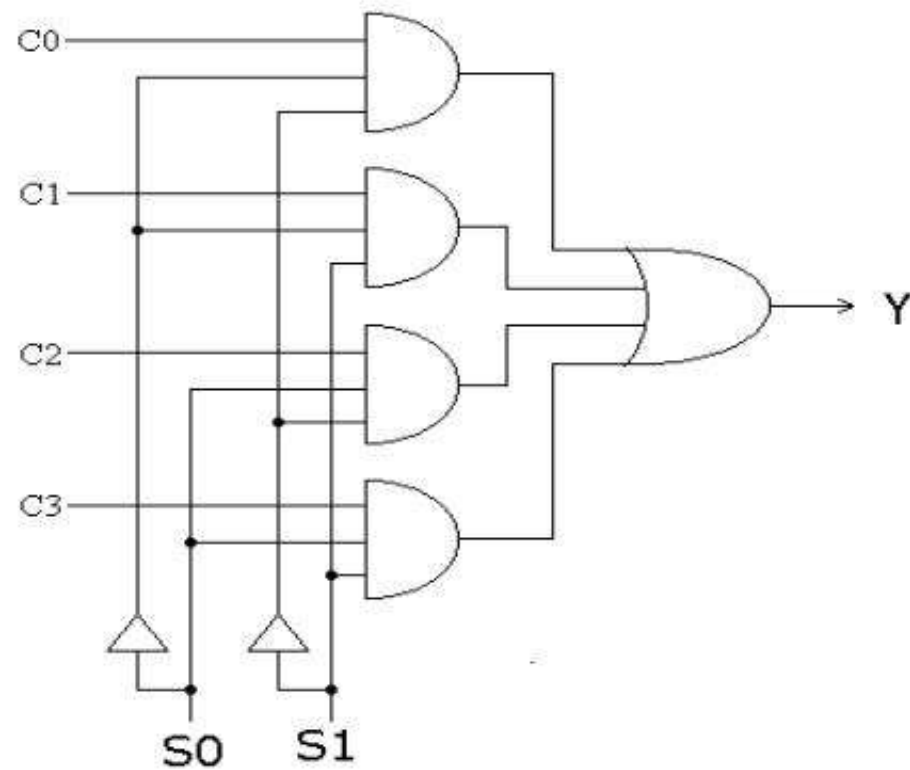
Block diagram of Multiplexer



4 to 1 Multiplexer Logic Symbol



4 to 1 Multiplexer Logic diagram



Cont.,

- To demonstrate the operation of the circuit, consider the case when $S_1 S_0 = 00$.
- If $S_1 S_0 = 00$ is applied to the select lines, the AND gate associated with D_0 will have two of its input equal to 1 and the third input connected to D_0 .
- The other three AND gates have 0 in at least one of their inputs, which make their output equal to 0.
- Hence, the OR output (Y) is equal to the value of D_0 .

Cont.,

- Thus, it provides a path from the selected input and the data on the input D_0 appears on the data-output line (Y).
- If $S_1 S_0 = 01$ (binary 1) applied to the select lines , the data on the input D_1 appears on the data output line.

Cont.,

- If $S_1 S_0 = 10$ (binary 2) is applied, the data on the input D_2 appears on the output line (Y).
- Similarly, if $S_1 S_0 = 11$ is applied, the data on D_3 is switched to the output line (Y).
- The AND gates and the inverters resemble a decoder circuit, and indeed they decode the input select lines.
- In general, a 2^n -to-1 multiplexer is constructed from n -to- 2^n decoder by adding to it 2^n input lines, one each AND gate.

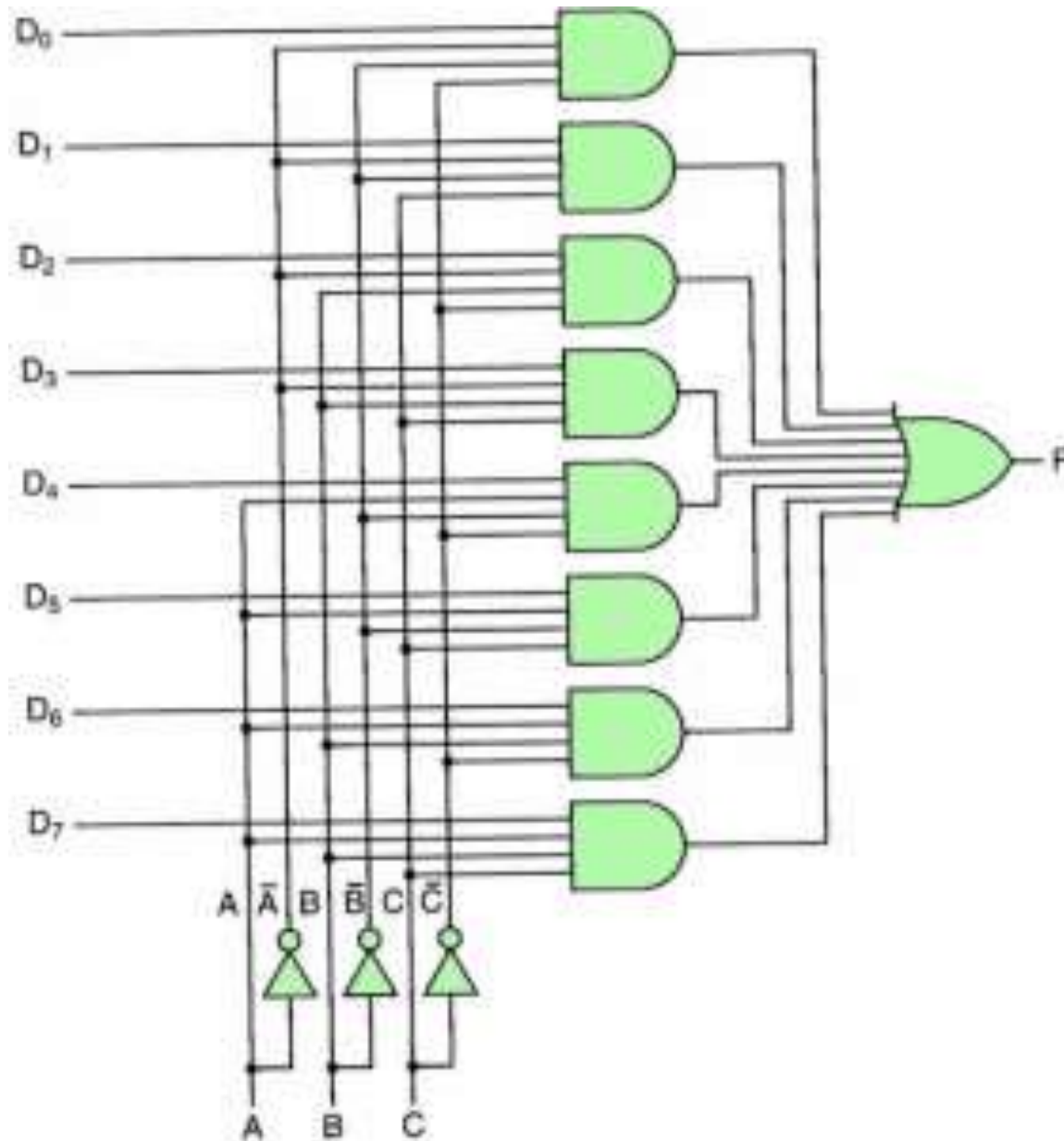
Cont.,

- The output of the AND gates are applied to a single OR gate to provide a single output.
- The size of the multiplexer is specified by the number 2^n of input lines and the single output line.
- Multiplexer ICs have an enable input to control the operation of the unit.
- The enable input (also called strobe) can be used to cascade two or more multiplexer ICs to construct a multiplexer with larger number of inputs

Truth table of 4-to-1 Multiplexer

Data select inputs		Output
S ₁	S ₀	Y
0	0	D ₀
0	1	D ₁
1	0	D ₂
1	1	D ₃

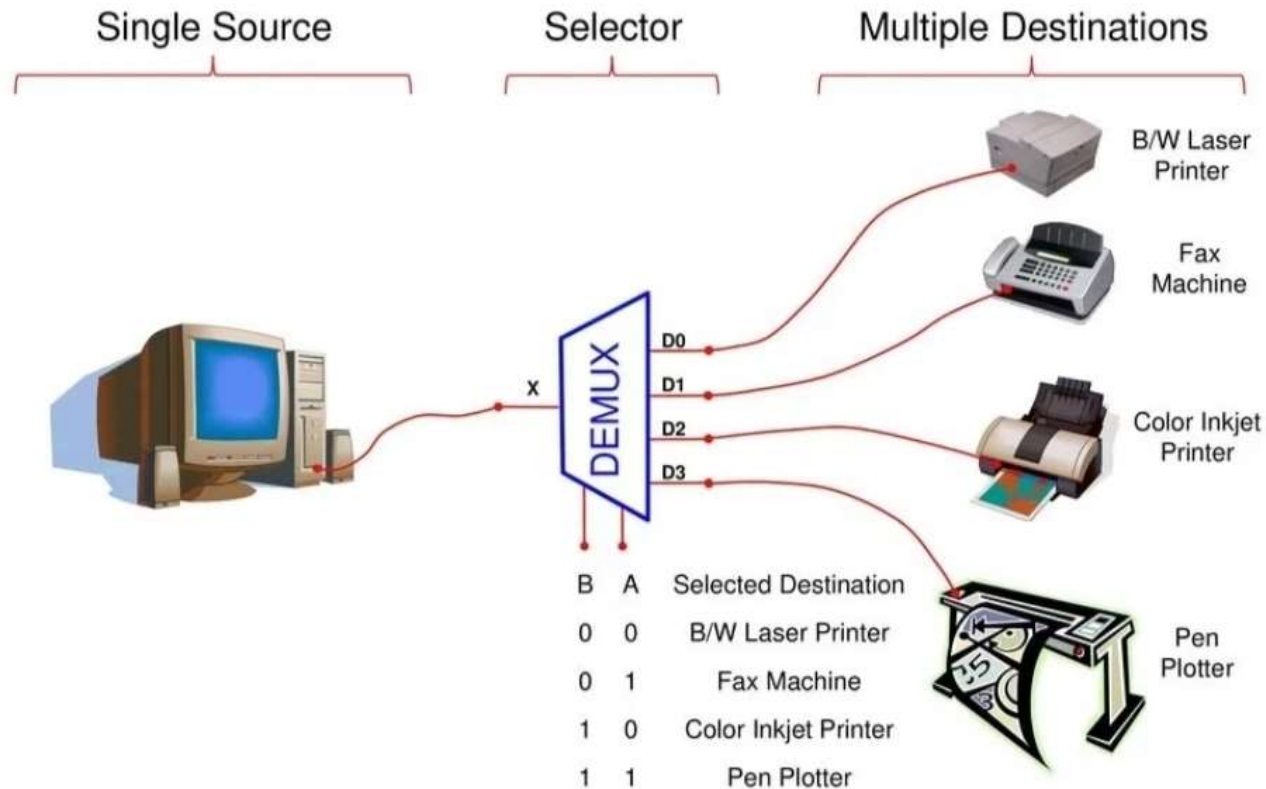
8-1 Multiplexer Circuit



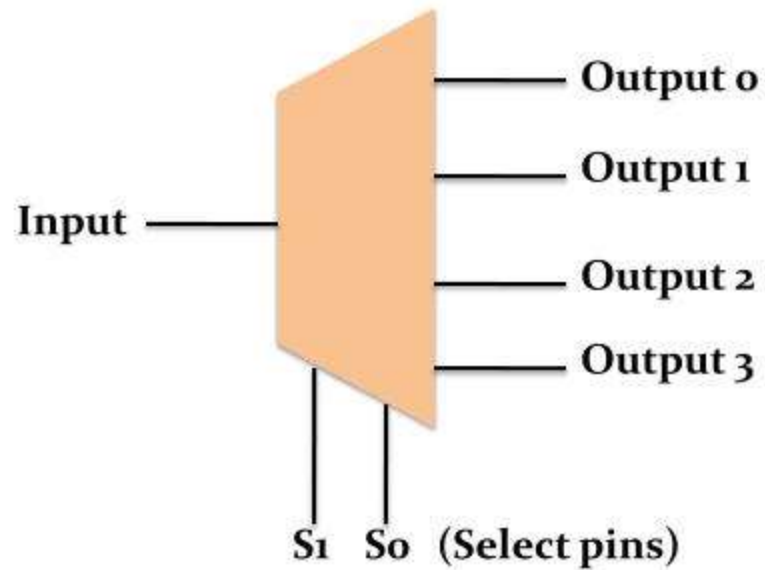
De-Multiplexer

- The De-Multiplexer is a combinational logic circuit that performs the reverse operation of multiplexer (Several output lines, one input line).
- De -Multiplexer means one to many. A De-Multiplexer is a circuit with one input and many output. By applying control signal, we can steer any input to the output. Few types of De -Multiplexer are 1-to 2, 1-to-4, 1-to-8 and 1-to 16 De -Multiplexer .
- De-Multiplexer is the process of taking information from one input and transmitting the same over one of several outputs.

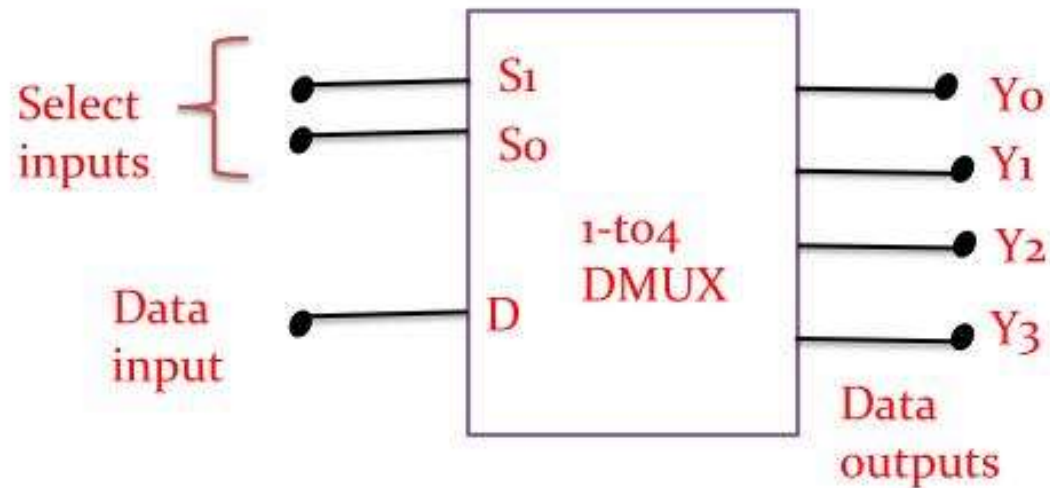
Typical Application of a DEMUX



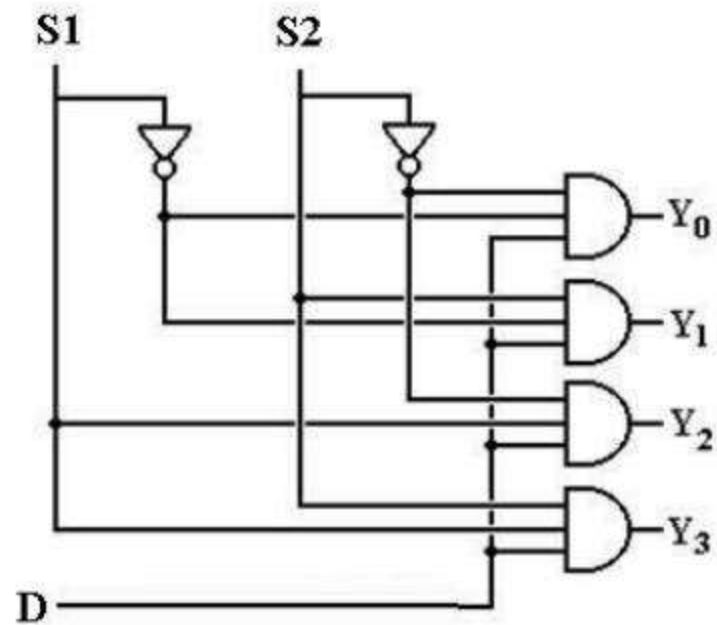
Block diagram of DeMultiplexer



1 to 4 DeMultiplexer Logic Symbol



1 to 4 DeMultiplexer Logic diagram



Truth Table of 1 to 4 DeMultiplexer

Data Input	Select Inputs		Output			
D	S ₁	S ₀	Y ₃	Y ₂	Y ₁	Y ₀
D	0	0	0	0	0	D
D	0	1	0	0	D	0
D	1	0	0	D	0	0
D	1	1	D	0	0	0

Applications of Multiplexer

Communication system – Communication system is a set of system that enable communication like transmission system, relay and tributary station, and communication network. The efficiency of communication system can be increased considerably using multiplexer. Multiplexer allow the process of transmitting different type of data such as audio, video at the same time using a single transmission line.

Computer Memory – A Multiplexer is used in computer memory to keep up a vast amount of memory in the computers, and also to decrease the number of copper lines necessary to connect the memory to other parts of the computer.

Applications of De-Multiplexer

Communication System - Communication system use multiplexer to carry multiple data like audio, video and other form of data using a single line for transmission. This process make the transmission easier. The demultiplexer receive the output signals of the multiplexer and converts them back to the original form of the data at the receiving end. The multiplexer and demultiplexer work together to carry out the process of transmission and reception of data in communication system.

THANK YOU